

Nuclear Energy: Energy Production in 5 EU Countries from 1965 to 2025

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Introduction

Energy production in the European Union (EU) is a current topic as countries address their own energy demands in different manners. Overall, energy production in the EU is a combination of different sources, such as, renewables which compile 46% of total EU energy production, followed by nuclear at 29%, solid fuels at 17%, natural gas at 5% and crude oil at 3% [1]. Nuclear energy is a unique source to investigate, considering the stigma and varied opinions EU countries have had over the years.

Thus, in this report the following research question is asked: *How has nuclear energy (electricity) production in Belgium, France, Germany, Italy and the Netherlands changed from 1965 to 2025?*

To answer this question a dataset from Our World in Data, which shows global electricity generation from nuclear (measured in terawatt-hours) between 1900 and 2025 will be used [2].

Data sample

The dataset [2] is divided into 4 columns:

- **Entity**: the name of the country
- **Code**: 3 letter country codes (most use ISO alpha-3)
- **Year**: time point
- **Nuclear**: electricity produced in terawatt-hours

Since, the dataset is global it was cleaned to only include data from Belgium, France, Germany, Italy and the Netherlands. Additionally, the data before 1965 is very limited, hence anything before that has also been removed.

Results and discussion

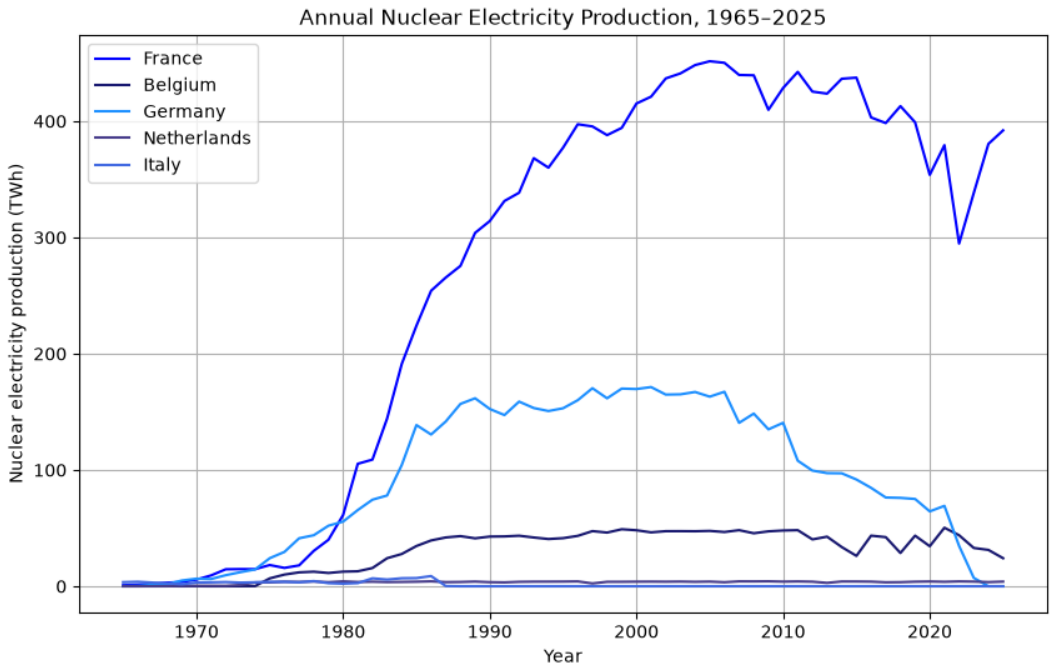


Figure 1: Line graph showing annual nuclear energy production of France, Belgium, Germany, Netherlands, and Italy between 1965 and 2025

Firstly, the line graph as seen in Figure 1 clearly shows that France has greatly increased electricity generation from nuclear power plants between mid-1970 up until 2025, with its peaks between 2000 and 2010. The second country with a largest production of electricity from nuclear is Germany, which had a stable period between mid-1980 and 2010, after which its a slow decrease until right after 2020 when the production becomes 0. Belgium, has one of the most stable productions between 1980 and mid-2010. The Netherlands, overall production is substantially smaller compared to the aforementioned countries. Lastly, Italy had a small spike in production in the 1980's after which it dropped to 0.

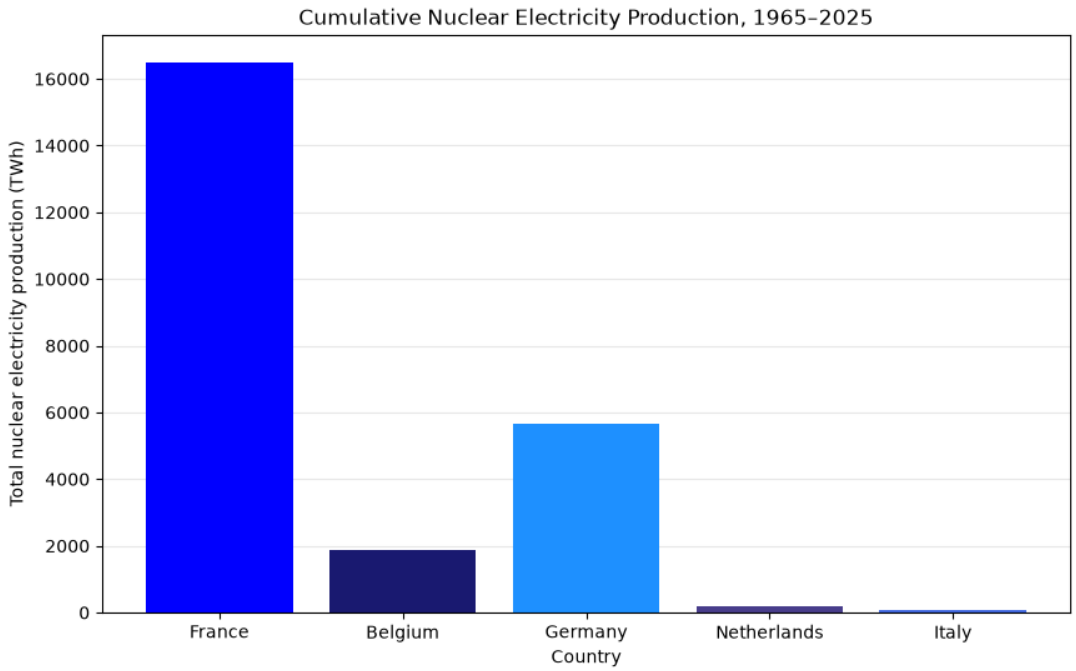


Figure 2: Bar graph showing cumulative energy production of France, Belgium, Germany, Netherlands, and Italy from 1965 to 2025

The second graph, as seen in Figure 2, shows the cumulative energy production of the five countries over a period of 60 years. France has the highest cumulative energy production, over 16000 TWh, followed by Germany even though production has been terminated. Belgium's production is just under 2000, meanwhile the Netherlands and Italy are close to 0.

Overall, it is clear that France has had the most electricity produced via nuclear energy, followed by Germany, even though it recently stopped, after which is Belgium, the Netherlands, and lastly Italy. Both graphs show that electricity produced by nuclear energy has varied between 1965 and 2025. This could be due to a variety of causes such as changes in the economy, politics, as well as, transitioning to other energy sources. Additionally, a potential extension of this project could be including and analyzing the land mass of the countries, to determine if that could influence electrcity production via nuclear.

References

[1] EUROSTAT *Energy production in the EU*. Available at: https://ec.europa.eu/eurostat/web/interactive-publications/energy-2025.

[2] Our world in data. *Electricity generated from nuclear energy dataset*. Available at: https://ourworldindata.org/grapher/electricity-mix?tab=table&source=nuclear&metric=generation&frequency=annual.